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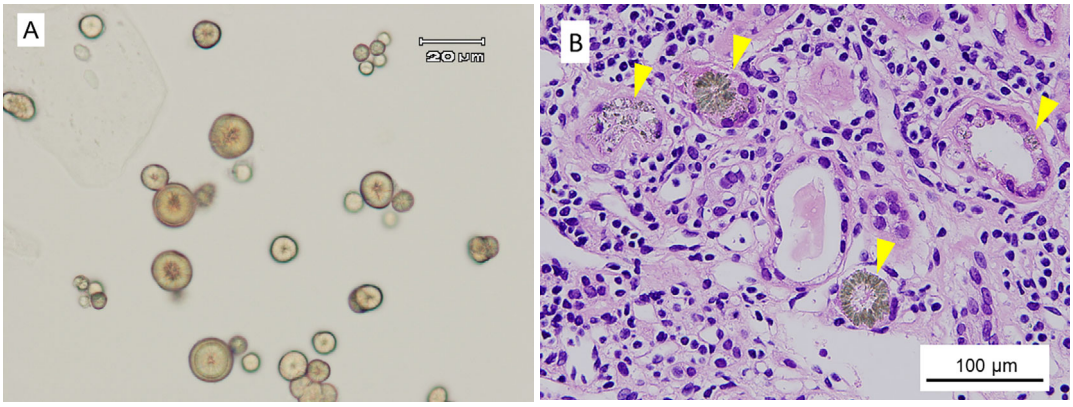
【 PICTURES IN CLINICAL MEDICINE 】

Urinary Crystals with 2,8-Dihydroxyadeninuria

Shinichi Mizuno

Key words: adenine phosphoribosyltransferase deficiency, crystal, 2,8-dihydroxyadenine, febuxostat, tubulointerstitial nephritis

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Picture.

A 51-year-old woman was admitted to our hospital after her serum creatinine level increased 0.6 mg/dL to 2.6 mg/dL over one year. She had no history of urolithiasis. A urinalysis showed characteristic annular crystals with central spicules (Picture A). Neither proteinuria nor hematuria was observed. A renal biopsy revealed tubulointerstitial nephritis associated with the crystals of various sizes inside the tubular lumen and tubular epithelium cytoplasm (Picture B, Hematoxylin and Eosin staining). A homozygous mutation [*APRT*J/APRT*J* (Met136Thr)] of the adenine phosphoribosyltransferase (APRT) gene was identified by a genetic analysis. Additionally, her parents and grandparents nonconsanguineous; however, her brother had been noted to have same urinary crystals. APRT deficiency is an autosomal recessive inheritance disorder and is a typical cause of urolithiasis, especially in childhood; however, individuals with APRT deficiency are occasionally asymptomatic until adulthood. The lack of APRT activity leads to impaired purine salvage metabolism, and causes the excessive production of urine insoluble 2,8-dihydroxyadenine (DHA) by xanthine oxidase (XO). Treatment with febuxostat (induction, 30 mg; maintenance, 10 mg), an XO inhibitor, resulted in the rapid disappearance of 2,8-DHA crystals in the patient's urine, and improved her renal function.

The author states that he has no Conflict of Interest (COI).

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Department of Nephrology, Japan Community Health Care Organization Sendai Hospital, Japan
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Correspondence to Dr. Shinichi Mizuno, s.mizuno@sendai-kidney.jp
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